

Golf's Golden Ratio – A Statistical Analysis of Tempo & Driving Distance

Section 1: Introduction

The significance of distance in golf is underscored by the R&A and USGA's¹ antagonistic decision to implement additional restrictions on the golf ball in 2028, commonly referred to as the “rollback”. This decision reflects the sport's recognition of the pivotal role that driving distance is having in shaping the professional game. In the 1950s, male professional golfers averaged distances ranging from 240 to 260 yards; fast forward to 2023, this average now hovers around 300 to 310, highlighting the general thirst for distance in professional golf (R.H. Sport, 2023).

The rationale behind this distance craving is relatively straightforward: the further you hit your tee shot, the closer to the pin the subsequent approach shot is. Proximity to the pin translates to increased accuracy. If a given player is already hitting the ball impressive distances, this probably further translates to needing to hit less demanding clubs irrespective to their distance to the flag, leading to even more precision. Greater accuracy results in fewer shots which leads to better scores.

In attempting to understand how to hit the ball further and more consistently, golf players resort to swing coaches and general literature on the mechanics of the golf swing. It is often considered an unspoken truth that good “tempo” (ratio between backswing downswing time) is synonymous with being a consistently performing golf player – but what does this mean? Until recently, a complete mystery, it was not until *Tour Tempo: Golf's Last Secret Finally Revealed*, that professionals tried to dispel the secrets of the elites' “perfect” tempo. In their study they revealed that most tour-level players have a 3:1 ratio – that is, they take three times longer to bring the club back than to get it from the top to the impact position (Novosel & Garrity, 2004). Given that these claims have since been uncontested, it would make sense that if professional golf players have the same tempo, what dictates how far they hit the ball, is how fast they perform their backswing since that in turn would determine how fast they complete the downswing and therefore how much kinetic energy their club transfers to the ball upon impact.

This analysis will attempt to test the mentioned hypothesis by expanding on the value of this discovery – we will assess the relationship between time taken to perform the backswing and driving

¹ R&A and USGA: The Royal and Ancient Golf Club of St. Andrews and the United States Golf Association are the primary governing bodies responsible for establishing and maintaining the rules of golf on an international level. They're crucial in shaping the sport's standards, organising major championships, and promoting the growth and integrity of the sport globally.

distance, and implicitly be exploring the relationship between tempo and distance. We have decided to focus on female golf players to expand the limited literature on female professional athletes and the attributes that go into their success. In this practice of inference, we will compare the driving distances of LPGA² golf players with the duration of their backswings and will also compare how big this effect is relative to alternative explanatory variables such as height. We will attempt to determine statistically whether there exists a relationship between the time to complete the backswing and driving distance.

Section 2: Methods, data collection

To collect a sample for this analysis we first accessed the LPGA's 2023 average driving distance leaderboard. This scoreboard compiles the average yardage of drives hit by members of the LPGA tour at events hosted by said organisation from the beginning of the 2023 season until the date this data was collected, the 8th of December 2023. We then generated a random sample of size 40 (a limited size due to the laborious nature of manual data collection) from this data frame and ensured that data points were collected from as wide a range of distances as possible – spread out data makes graphic representations of linear regression models more apparent.

Once selected, we proceeded to search online on YouTube for the most recent tournament footage of the selected players hitting their drivers at real speed. To reduce the human error when recording the duration of their backswings, we played the videos at a tenth of the original speed and using a stopwatch with millisecond accuracy recorded the time it took from the moment the club head started moving to when it reached the top of the backswing. Once again to reduce the human error in these measurements, we proceeded to repeat this 3 times for every video and took an average of our measurements. Once we collected these averages, we divided these time measurements by 10 to compute the real-time backswing duration in seconds which we paired with the average distance in yards hit by these players. From a variety of sources, we found the heights of all these players in feet and inches and proceeded to convert and pair these measurements to the original data in centimetres. Note that these are all continuous numeric variables – we therefore decide to use a linear regression model to assess the potential relationship between these variables in the following section.

² The LPGA, or Ladies Professional Golf Association is a professional organisation for female golfers best known as the governing body for women's professional golf, organising tournaments and promoting the sport worldwide

Section 3: Data Analysis

We are concerned with what the duration of a player's backswing (explanatory variable in this case) may say of their average driving distance (response variable in this analysis). We can attempt to model this relationship using a simple linear regression model described as follows:

$$y = \beta_0 + \beta_1 x + \epsilon$$

This model argues that the average driving distance of a player (y) is a linear function of the duration of said player's backswing (x) with intercept β_0 and slope β_1 , plus noise that keeps the data from being colinear. More specifically ϵ , represents the error term, accounting for variability in y that isn't explained by the linear relationship with x : factors that aren't included in the model, measurement errors, or inherent randomness in the relationship between the variables. In a world where there is no relationship between backswing duration and height, we anticipate the best-fitting least squares regression line to be flat, that is $H_0: \beta_1 = 0$, however we argue $H_A: \beta_1 \neq 0$. Using R, we model the following least squares regression line (Fig. 1). The regression line generated from our data isn't flat which seems promising – the negative slope seems to suggest that the slower your backswing, the shorter you hit the ball.

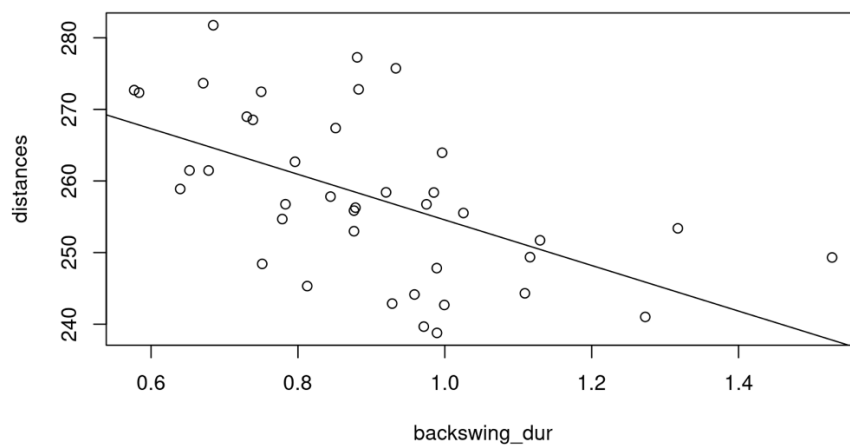


Fig. 1

In this data's noise (ϵ) we can easily identify potential factors linked to backswing duration and average driving distances that aren't being taken into consideration such as players' stature. Height could easily be a factor in determining how far a player hits a ball influencing the biomechanics of a golf swing – taller players may have longer limbs, potentially affecting their range of motion during a backswing. We have access to this data and therefore decide to carry out this regression with two variables: x_1 , duration of

player's backswing, and x_2 , said player's height, to better identify just how significant of a factor backswing duration is in determining how far they hit the ball. Bringing in this variable can help us reduce the variability observed with our data. We now want to model this regression as: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$ and perform multiple linear regression, evaluating a least squares regression plane attempting to minimise $Q(\beta_0, \beta_1, \beta_2) = \sum_{i=1}^{40} (y_i - \beta_0 - \beta_1 x_{1i} - \beta_2 x_{2i})^2$ – reducing the amount of squared residuals (how much we under/overfit the data by using our explanatory variables). Using R, our data yields $\hat{\beta}_0 = 259.8$, $\hat{\beta}_1 = -31.4$ and $\hat{\beta}_2 = 0.154$. That is:

$$\hat{y} = 259 - 31.3 \hat{x}_1 + 0.154 \hat{x}_2$$

This model suggests that using our data, we can predict that for every second increase in the backswing time, the estimated average driving distance of LPGA players decreases by approximately 31.4 yards when holding height constant (for a golfer, a potential gain of 30 yards off the tee could be incredibly beneficial). On the other hand, for every centimetre increase in height, the estimate average driving distance increases approximately 0.154 yards holding backswing duration constant (this change is insignificant).

There are certain assumptions about the residuals of our model that are made when performing linear regression that we must address before drawing any conclusions from this data: linearity, homoskedasticity and normality.

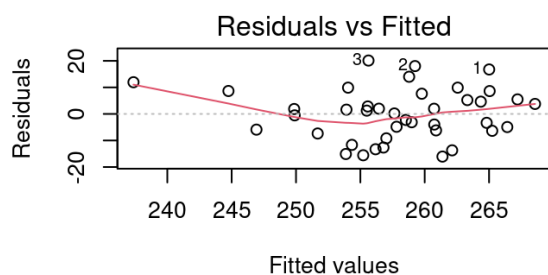


Fig. 2

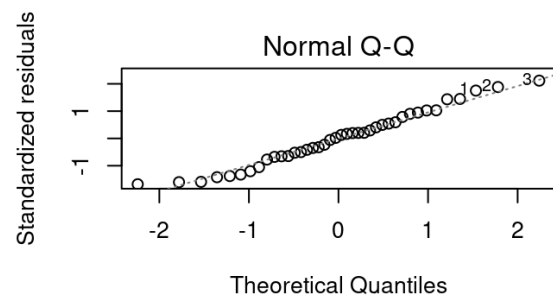


Fig. 3

In Fig. 2, we can observe that despite having a one greatly outlying residual below 240 yards average distance, quite generally, the data points are evenly distributed at either side of the residuals = 0 line which implies both linearity and constant variance. Similarly, our Normal Q-Q plot, Fig. 3, shows that our standardised residuals follow the normal distribution quite closely, and with our sample size the

Central Limit Theorem seems to apply fast enough. Therefore, it follows that our assumptions for this type of regression are met.

Looking back at our model, $\hat{y} = 259 - 31.3 \hat{x}_1 + 0.154 \hat{x}_2$, we compute the t-statistic for our $\hat{\beta}_1 = 31.3$ to be -3.952. This value gives an indication as to whether the estimated coefficient for backswing duration deviates significantly from 0. The p-value is of $0.000336 < \alpha = 0.05$ which indicates that backswing duration is statistically significant in predicting average driving distance. The chance that this data emerges in a world where there is no relationship between the two variables is close to 0 and therefore implies that there exists a relationship between the duration of the backswing and the average drive length of an LPGA player. We reject the null hypothesis since it seems likely that the duration of backswing gives us valuable information to predict average driving length!

On the other hand, the t-statistic for $\hat{\beta}_2 = 0.154$, is 0.497, which being very close to 0, has a p-value of $0.622404 > \alpha = 0.05$. This data is likely to occur in a world where driving distance doesn't have a special relationship with player's height. It suggests that height is not statistically significant in predicting the distance when keeping things like backswing duration constant. Our multiple r-squared value of 0.3083 implies a moderate correlation between our variables – close to a third of the variation of average driving distance for LPGA players can be explained by the time it takes to complete the backswing and height.

Section 4: Conclusion

Our statistical analysis of the relationship between backswing duration and average driving distance reveals that there's likely a negative correlation between the two at least when it comes to professional female golfers – this implies that the faster you set up the golf club at the top of the swing, the faster you bring it down, an argument that lies consistently with the belief that professional golfers have a somewhat similar tempo. How far they hit it can be better predicted by their backswing time than by their height, which when thinking about biomechanics of golf is a surprising finding. Ideally this experiment could be performed on a larger sample size and on male professional golfers too. It would also be interesting to explore this relationship as it pertains to golfers of different levels since the findings could potentially reveal a different approach to teaching and levelling up in the sport.

Sources and References

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